

Multiple Websites Deployment on NGINX Server on Ubuntu Instance

This document provides a step-by-step guide to deploying five separate websites simultaneously on a single NGINX server using an Ubuntu instance.

1. Introduction to NGINX

NGINX (pronounced “engine x”) is an open-source HTTP web server and reverse proxy known for its high performance, stability, and low resource utilization.

2. Installing NGINX

To begin, update your package list and install NGINX.

```
sudo apt-get update -y
sudo apt-get install nginx -y
```

Verify Installation

Check the status and ensure NGINX is running and enabled to start on boot.

```
systemctl status nginx
systemctl start nginx
systemctl enable nginx
nginx -v
```

By default, NGINX runs on port 80. You can verify this by accessing the public IP of your instance.

3. Preparing Website Directories

Navigate to the web root directory and create separate folders for each of the five sites.

```
cd /var/www/  
sudo mkdir site1 site2 site3 site4 site5
```

Creating Index Files

Create a basic `index.html` file for each site.

For Site 1:

```
cd /var/www/site1  
sudo vi index.html  
# Add content and save (Esc :wq)
```

(Repeat the process for site2, site3, site4, and site5)

Directory Structure Verification

You can use the `tree` command to verify the structure.

```
sudo apt install tree -y  
tree /var/www/
```

4. Configuring NGINX Server Blocks

NGINX configurations are managed in the `/etc/nginx/sites-available/` directory.

Default Configuration Reference

The default configuration file can be used as a template.

```
cd /etc/nginx/sites-available/  
cat default
```

Creating Site-Specific Configurations

Create separate configuration files for each site, assigning them unique ports.

Configuration for Site 1 (config-1)

```
server {  
    listen 210;  
    server_name site1.com www.site1.com;  
    root /var/www/site1;  
    index index.html;  
    location / {  
        try_files $uri $uri/ =404;  
    }  
}
```

Configuration for Site 2 (config-2)

```
server {  
    listen 220;  
    server_name site2.com www.site2.com;  
    root /var/www/site2;  
    index index.html;  
    location / {  
        try_files $uri $uri/ =404;  
    }  
}
```

Configuration for Site 3 (config-3)

```
server {
    listen 230;
    server_name site3.com www.site3.com;
    root /var/www/site3;
    index index.html;
    location / {
        try_files $uri $uri/ =404;
    }
}
```

Configuration for Site 4 (config-4)

```
server {
    listen 240;
    server_name site4.com www.site4.com;
    root /var/www/site4;
    index index.html;
    location / {
        try_files $uri $uri/ =404;
    }
}
```

Configuration for Site 5 (config-5)

```
server {
    listen 250;
    server_name site5.com www.site5.com;
    root /var/www/site5;
    index index.html;
    location / {
        try_files $uri $uri/ =404;
    }
}
```

Note: Do not use port 110 as it is reserved for the Post Office Protocol (POP3).

5. Enabling the Configurations

To enable the sites, create symbolic links from the `sites-available` directory to the `sites-enabled` directory.

```
sudo ln -s /etc/nginx/sites-available/config-1 /etc/nginx/sites-enabled/  
sudo ln -s /etc/nginx/sites-available/config-2 /etc/nginx/sites-enabled/  
sudo ln -s /etc/nginx/sites-available/config-3 /etc/nginx/sites-enabled/  
sudo ln -s /etc/nginx/sites-available/config-4 /etc/nginx/sites-enabled/  
sudo ln -s /etc/nginx/sites-available/config-5 /etc/nginx/sites-enabled/
```

Applying Changes

Restart or reload NGINX to apply the new configurations.

```
sudo systemctl restart nginx  
sudo systemctl reload nginx
```

6. Security Group Configuration

In your cloud provider's console (e.g., AWS), update the inbound security rules to allow traffic on the specified ports.

1. Go to **Security -> Security Groups**.
2. Select **Edit inbound rules**.
3. Add a rule:
 - **Type:** Custom TCP
 - **Port Range:** 210-250
 - **Source:** Anywhere (0.0.0.0/0)
4. Save the rules.

7. Launching and Verifying the Sites

You can now access your websites using the public IP address of your instance followed by the assigned port number.

- **Default Site:** `http://[public-ip]/`
- **Site 1:** `http://[public-ip]:210/`
- **Site 2:** `http://[public-ip]:220/`
- **Site 3:** `http://[public-ip]:230/`
- **Site 4:** `http://[public-ip]:240/`
- **Site 5:** `http://[public-ip]:250/`